

Information-Theoretic Observation Geometry

Typed information loss, data processing, and recoverable invariants

Stassis Research Program

Strict GO Core reference revision v0.2 — 2026

Status. This revision is the information-theoretic P1 reference module for GO Core v0.2. It fixes the logarithm base, separates raw and normalized defects, records the degenerate-source zero policy, types the full observation protocol, and distinguishes probabilistic information loss from metric covering loss. All theorem statements below are standard finite-alphabet information theory unless explicitly marked as a model or interface proposition.

Abstract

An observation is represented by a reduction, a deterministic readout, and a stochastic channel, rather than by an unspecified map. For finite hidden and observed alphabets and a declared logarithm base $b > 1$, the raw information defect is the conditional entropy $\mathcal{D}_{\lambda, b}^{\text{info}}(P_X) = H_b(X | Y_\lambda)$. It depends on both the channel and the source law. Its normalized form divides by $H_b(X)$ with an explicit zero-source policy and is invariant under a change of information unit. Deterministic channels admit an exact fiber formula; post-processing monotonically decreases retained mutual information and increases the defect; Kullback-Leibler distinguishability contracts. Exact recovery of a finite target $Q = f(X)$ is equivalent to $H_b(Q | Y) = 0$, while Fano's inequality supplies a one-sided obstruction to approximate recovery. Continuous variables are handled through relative entropy, mutual information, and declared quantizers, not through coordinate-dependent differential entropy. The document supplies a typed interface and claim audit; it does not claim new information-theoretic inequalities or a derivation of physical dynamics.

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1 Scope, conventions, and claim boundary

Fix a logarithm base $b > 1$. Unless a standard-Borel extension is stated explicitly, every alphabet is finite and every random variable is defined on a common probability space. Probability laws are denoted by P_X , P_{XY} , and so forth. We use

$$0 \log_b 0 := 0, \quad \log_b \frac{p}{0} := +\infty \quad (p > 0).$$

Information quantities are physically dimensionless but have the semantic type information; their numerical unit is fixed by b (bits for $b = 2$, nats for $b = e$).

Boundary 1.1 (Claim boundary). Shannon entropy is not identified with thermodynamic entropy without a separate physical model. An information channel does not create a hidden state, and information theory alone does not derive spacetime, gravity, mass, charge, or consciousness. The contribution of this note is a type-safe interface between standard information theory and the Geometry of Observation protocol.

1.1 The GO observation protocol

For protocol $\lambda \in \Lambda$, write

$$X \xrightarrow{R_\lambda} X_\lambda \xrightarrow{O_\lambda} U_\lambda \xrightarrow{K_\lambda} Y_\lambda \xrightarrow{\hat{q}_\lambda} \hat{Q}_\lambda.$$

Here R_λ is a model reduction, O_λ is an ideal deterministic readout, and $K_\lambda(y | u)$ is a Markov kernel. The induced hidden-to-data channel is

$$W_\lambda(y | x) = K_\lambda(y | O_\lambda(R_\lambda x)). \quad (1)$$

If the reduction itself is stochastic, it is absorbed into W_λ by kernel composition. A frame or coordinate transformation is instead an invertible representation map and is not counted as information loss.

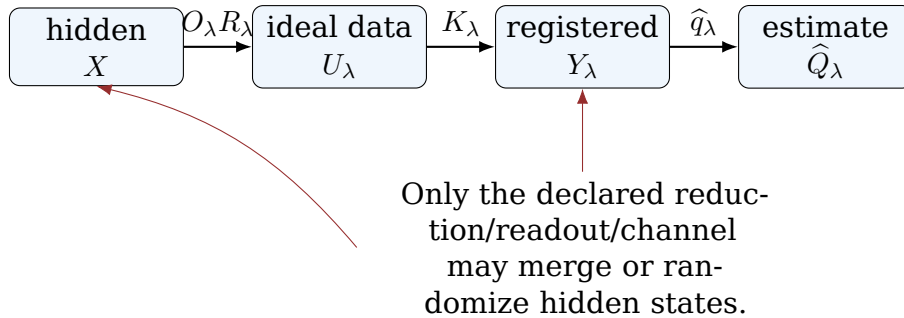


Figure 1: Typed observation chain. A channel is distinct from a change of coordinates or frame.

2 Information quantities and change of unit

Definition 2.1 (Entropy, conditional entropy, and mutual information). For a finite random variable X ,

$$H_b(X) = - \sum_{x \in \mathcal{X}} P_X(x) \log_b P_X(x).$$

For a pair (X, Y) ,

$$H_b(X | Y) = \sum_{y \in \mathcal{Y}} P_Y(y) H_b(X | Y = y),$$

and

$$I_b(X; Y) = H_b(X) - H_b(X | Y) = H_b(Y) - H_b(Y | X).$$

Definition 2.2 (Relative entropy). For laws P, Q on a finite set,

$$D_{\text{KL},b}(P \| Q) = \sum_x P(x) \log_b \frac{P(x)}{Q(x)} \in [0, +\infty].$$

It is finite precisely when $P \ll Q$ in the finite setting.

Theorem 2.3 (Information-unit change). *Let $b, c > 1$. Every entropy, conditional entropy, mutual information, and finite relative entropy in base b is related to its base- c value by*

$$H_b = \log_b(c) H_c, \quad I_b = \log_b(c) I_c, \quad D_{\text{KL},b} = \log_b(c) D_{\text{KL},c}.$$

Consequently, any ratio of two finite positive information quantities computed with the same base is independent of the base.

Proof. The identity $\log_b t = \log_b(c) \log_c t$ holds for $t > 0$. Substitution in each finite sum gives the first three relations. The common positive factor cancels in a ratio. $\square$

Remark 2.4 (Semantic units). The theorem is the information analogue of a coherent physical unit change. Subtracting a bit-valued entropy from a nat-valued entropy without conversion is forbidden, even though both are physically dimensionless.

3 Observation defect and deterministic fibers

Definition 3.1 (Raw information defect). For source law P_X and channel W_λ producing Y_λ , define

$$\mathcal{D}_{\lambda,b}^{\text{info}}(P_X) := H_b(X | Y_\lambda).$$

This is a function of the pair (P_X, W_λ) ; it is not an intrinsic scalar of the channel alone.

Definition 3.2 (Normalized information defect). The normalized defect is

$$\delta_\lambda^{\text{info}}(P_X) = \begin{cases} \frac{H_b(X | Y_\lambda)}{H_b(X)}, & H_b(X) > 0, \\ 0, & H_b(X) = 0. \end{cases}$$

The second line is the declared *degenerate-source zero policy*: a source supported at one state contains no source distinguishability to lose.

Proposition 3.3 (Decomposition, bounds, and unit invariance). *For every finite source and channel,*

$$H_b(X) = I_b(X; Y_\lambda) + \mathcal{D}_{\lambda,b}^{\text{info}}(P_X), \quad 0 \leq \mathcal{D}_{\lambda,b}^{\text{info}}(P_X) \leq H_b(X).$$

Moreover $0 \leq \delta_\lambda^{\text{info}}(P_X) \leq 1$, and $\delta_\lambda^{\text{info}}(P_X)$ is independent of b .

Proof. The identity is the definition of mutual information. Nonnegativity of conditional entropy and mutual information gives the bounds. The normalized claims follow from theorem 2.3 and the zero policy in definition 3.2. $\square$

Theorem 3.4 (Deterministic decomposition and fiber formula). *Let $Y = O(X)$ for a deterministic map $O : \mathcal{X} \rightarrow \mathcal{Y}$. Then*

$$H_b(X) = H_b(Y) + H_b(X | Y), \quad I_b(X; Y) = H_b(Y),$$

and

$$H_b(X | Y) = \sum_{y: P_Y(y) > 0} P_Y(y) \left[- \sum_{x \in O^{-1}(y)} P_{X|Y}(x | y) \log_b P_{X|Y}(x | y) \right]. \quad (2)$$

The defect vanishes if and only if X is a function of Y almost surely. Equivalently, every fiber meeting $\text{supp } P_X$ contains exactly one point of positive conditional probability.

Proof. Since Y is a function of X , $H_b(Y | X) = 0$. Apply the chain rule to $H_b(X, Y)$ in both orders. Expanding the definition of conditional entropy gives (2). A finite conditional entropy is zero exactly when every positive-probability conditional law is a point mass. $\square$

4 Post-processing, preorder, and divergence contraction

Definition 4.1 (GO post-processing preorder). For two channels with common input alphabet, write

$$W_1 \preceq_{\text{GO}} W_2 \iff W_2 = T \circ W_1$$

for some Markov kernel T . Thus W_2 is a garbling of W_1 ; the left channel is at least as informative. This convention agrees with the deterministic relation-refinement order used in the GO core.

Theorem 4.2 (Data processing and defect monotonicity). *If $X \rightarrow Y \rightarrow Z$ is a Markov chain, then*

$$I_b(X; Z) \leq I_b(X; Y), \quad H_b(X | Z) \geq H_b(X | Y).$$

More exactly,

$$H_b(X | Z) - H_b(X | Y) = I_b(X; Y | Z) \geq 0.$$

Proof. The chain rule gives

$$I_b(X; Y, Z) = I_b(X; Z) + I_b(X; Y | Z)$$

and

$$I_b(X; Y, Z) = I_b(X; Y) + I_b(X; Z | Y).$$

The Markov property makes the last conditional mutual information zero. The stated identities follow. $\square$

Theorem 4.3 (Relative-entropy contraction). *For a Markov kernel $W : \mathcal{X} \rightsquigarrow \mathcal{Y}$ and laws P, Q on $\mathcal{X}$,*

$$D_{\text{KL}, b}(WP \| WQ) \leq D_{\text{KL}, b}(P \| Q).$$

Proof. Apply the log-sum inequality to $a_x = P(x)W(y | x)$ and $b_x = Q(x)W(y | x)$ for each y , then sum over y . Extended values are understood when absolute continuity fails. $\square$

Definition 4.4 (Channel-level KL contraction profile). For a fixed finite channel W , define

$$\eta_{\text{KL},b}(W) = \sup_{\substack{P,Q: \\ 0 < D_{\text{KL},b}(P\|Q) < \infty}} \frac{D_{\text{KL},b}(WP\|WQ)}{D_{\text{KL},b}(P\|Q)},$$

with value 0 if the admissible set is empty.

Proposition 4.5. $0 \leq \eta_{\text{KL},b}(W) \leq 1$, and the value is independent of b .

Proof. The bounds follow from theorem 4.3; base independence follows from the common factor in theorem 2.3. $\square$

5 Recoverable targets

Definition 5.1 (Exact recoverability). Let $Q = f(X)$ take values in a finite alphabet $\mathcal{Q}$. It is recoverable from Y if there is a decoder $g : \mathcal{Y} \rightarrow \mathcal{Q}$ such that $Q = g(Y)$ almost surely.

Theorem 5.2 (Entropy criterion for exact recoverability). For finite Q and Y ,

$$Q \text{ is recoverable from } Y \iff H_b(Q | Y) = 0.$$

If $Y = O(X)$ is deterministic, this is equivalent to f being constant on each fiber $O^{-1}(y) \cap \text{supp } P_X$, up to null states.

Proof. The conditional entropy vanishes precisely when each conditional law $P_{Q|Y=y}$ with $P_Y(y) > 0$ is a point mass. Selecting its unique atom defines $g(y)$. The deterministic fiber statement is the same condition written in terms of O . $\square$

Corollary 5.3 (Target recovery is weaker than state recovery). It is possible that

$$H_b(X | Y) > 0 \quad \text{but} \quad H_b(f(X) | Y) = 0.$$

Thus recovery claims must name the target; recovery of one invariant does not imply recovery of the hidden state.

Theorem 5.4 (Fano inequality). Let $|\mathcal{Q}| = M \geq 2$, let $\hat{Q} = g(Y)$ be any decoder, and put $P_e = P(\hat{Q} \neq Q)$. Then

$$H_b(Q | Y) \leq h_b(P_e) + P_e \log_b(M - 1),$$

where

$$h_b(p) = -p \log_b p - (1 - p) \log_b(1 - p).$$

Remark 5.5 (One-sided use). Fano's inequality obstructs very small error when the unresolved conditional entropy is large. Its converse use requires additional hypotheses or sharper reverse bounds. No claim that "small conditional entropy automatically supplies a stable physical inverse" is made here.

5.1 Target-normalized defect

For nonconstant $Q = f(X)$, define

$$\delta_{Q,\lambda}^{\text{info}}(P_X) = \frac{H_b(Q | Y_\lambda)}{H_b(Q)}.$$

If $H_b(Q) = 0$, the same degenerate-target zero policy assigns value 0. This target-specific defect is base invariant and lies in $[0, 1]$.

6 Calibrating examples

6.1 Finite parity aliasing

Let X be uniform on $\{0, 1, \dots, 2m - 1\}$ and let $Y = X \bmod 2$. Then

$$H_b(X) = \log_b(2m), \quad H_b(Y) = \log_b 2, \quad H_b(X | Y) = \log_b m.$$

The channel retains parity and loses the index within each parity fiber. For $b = 2$, the retained information is exactly one bit.

6.2 Binary symmetric readout

Let $X \sim \text{Bernoulli}(1/2)$ and $Y = X \oplus N$, where $N \sim \text{Bernoulli}(\alpha)$ is independent and $0 \leq \alpha \leq 1/2$. Then

$$H_2(X | Y) = h_2(\alpha), \quad I_2(X; Y) = 1 - h_2(\alpha), \quad \delta^{\text{info}} = h_2(\alpha).$$

This example separates deterministic fiber merging from stochastic sensor noise.

6.3 Quantized hidden circle

Let a hidden variable be uniform on a circle of circumference $2\pi r$. Choose a declared arc-length quantizer whose cells have diameter at most $\varepsilon_\ell > 0$. With

$$m_\varepsilon = \left\lceil \frac{2\pi r}{\varepsilon_\ell} \right\rceil$$

equal-probability arc cells, the unresolved discrete fiber entropy grows as

$$\log_b m_\varepsilon = \log_b \frac{r}{\varepsilon_\ell} + O(1).$$

The logarithm contains the dimensionless ratio r/ε_ℓ . This is a quantized probabilistic statement, not an entropy assigned directly to an unquantized continuum.

6.4 Conditional chromatic multiplicity

Let P be an observed cell and C a hidden color, phase, or label. Observation of P alone leaves defect

$$H_b(P, C | P) = H_b(C | P).$$

If $C = c(P)$ is a deterministic field, this quantity is exactly zero. Nonzero chromatic information loss requires several conditionally possible labels at at least one observed cell; a deterministic graph lift by itself does not create hidden entropy.

7 Continuous variables and declared quantizers

For random elements in standard Borel spaces, define mutual information by

$$I_b(X; Y) = D_{\text{KL}, b}(P_{XY} \| P_X \otimes P_Y),$$

possibly $+\infty$. This definition is invariant under measurable bijections of the alphabets. By contrast, if p_X is a density relative to a dimensional coordinate measure dx , a dimensionally admissible coordinate differential entropy requires a positive reference

volume v_* with the same units as dx :

$$h_{b,v_*}(X) = - \int p_X(x) \log_b(p_X(x)v_*) dx.$$

It depends on v_* and changes under coordinates; it can also be negative. The unreferenceed shorthand $-\int p \log p dx$ is prohibited by the GO logarithm rule. Differential entropy is therefore not used as a coordinate-free GO defect.

Let $q_\varepsilon : \mathcal{X} \rightarrow \{1, \dots, N_\varepsilon\}$ be an explicitly declared finite quantizer or measurable partition. The finite observable

$$X_\varepsilon = q_\varepsilon(X)$$

has the well-typed defect

$$H_b(X_\varepsilon | Y).$$

Every limit $\varepsilon \downarrow 0$ must state the family of quantizers, the metric and reference scale, and whether the limit is finite, divergent, or renormalized. Covering entropy constrains how many geometric cells are available, but it is not generally equal to a distribution-dependent conditional entropy. That distinction is developed in the companion Metric Entropy v0.2 module.

8 Observation design

Model 8.1 (Information bottleneck interface). Let $Q = f(X)$ be the target and require the Markov condition $Q - X - Y$. For a declared channel class $\mathcal{W}$ and information budget R_b in base- b units, one may study

$$\sup_{W \in \mathcal{W}} I_b(Q; Y) \quad \text{subject to} \quad I_b(X; Y) \leq R_b.$$

The channel class, source law, alphabet restrictions, and existence of an optimizer are part of the model, not automatic consequences of the notation.

Model 8.2 (Typed rate-distortion interface). Let d_Q have physical dimension $\mathbf{d}$ and choose a strictly positive reference distortion d_{ref} of the same dimension. The normalized loss

$$\bar{d}_Q(q, \hat{q}) = \frac{d_Q(q, \hat{q})}{d_{\text{ref}}}$$

is dimensionless. A rate-distortion problem may then constrain $\mathbb{E}[\bar{d}_Q] \leq D$ with dimensionless D . The source, admissible reproduction alphabet, and channel class must be declared.

9 GO Core quantity and protocol ledger

9.1 Map register

Map	Domain	Codomain	GO kind and loss status
R_λ	hidden state X	reduced state X_λ	reduction; loss possible
O_λ	X_λ	ideal datum U_λ	deterministic observation; loss possible
K_λ	U_λ	law on Y_λ	stochastic channel; loss possible
T	Y	law on Z	post-processing channel; loss possible
$\hat{q}_\lambda$	Y_λ	target estimate	estimator; loss is not a map attribute

9.2 Quantity register

Quantity	Physical dimension	Semantic type	Required protocol fields
$P_X(x), W(y x)$	0	probability	alphabet, normalization, source/channel law
$H_b, I_b, D_{\text{KL},b}$	0	information in base b	logarithm base and finiteness policy
$\mathcal{D}_{\lambda,b}^{\text{info}}$	0	raw information defect	source law, channel, target (if any)
$\delta_\lambda^{\text{info}}$	0	normalized loss	positive denominator or declared zero policy
d_Q	declared d	distortion	target space, metric/loss, reference distortion
$\bar{d}_Q$	0	normalized loss	$d_{\text{ref}} > 0$ with the same dimension

The complete protocol ledger declares:

1. hidden alphabet and source law;
2. system descriptor and protocol index λ ;
3. reduction, ideal observation, noise channel, and observed alphabet;
4. reference frame (or the explicit statement not applicable);
5. quantizer/partition and spatial/temporal resolution, including not used where appropriate;
6. observation horizon;
7. estimator or direct-readout declaration;
8. exactness, uncertainty, and missing-data policy;
9. neutral/SI unit context and information base;
10. raw defect, normalizer, and zero policy.

10 Claim audit and conclusion

Established under stated hypotheses	Not established by this module
Conditional entropy measures unresolved finite-source uncertainty for a declared law and channel.	An intrinsic, prior-free scalar loss for every observation device.
Post-processing decreases mutual information and contracts KL divergence.	Post-processing can reconstruct information absent from its input.
$H_b(Q Y) = 0$ characterizes exact finite-target recovery almost surely.	Recovery of one invariant implies recovery of the full hidden state.
Quantized continuous variables admit finite information defects.	Differential entropy is a coordinate-invariant hidden-state defect.
The normalized defect is invariant under information-unit changes.	Information defect is automatically thermodynamic entropy or physical energy.

The module closes a probabilistic interface:

$$\boxed{\text{declared source} + \text{declared channel} \mapsto (I_b(X; Y), H_b(X | Y), \eta_{\text{KL}}(W))}.$$

These are typed, noninterchangeable objects. The results are mathematically standard; the framework contribution is their integration into a reproducible observation protocol with explicit units, targets, normalizations, and claim status.

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